José Thiéry M. Hagbe

Mathematics and machine learning, Abomey-Calavi, Benin
jhagbe@africanschoolofeconomics.com • josehagbe3.github.io • ORCID 0000-0002-4189-5221 • github.com/josehagbe3

Research interests

Guarantees for safety-constrained prediction (recall floors, selective and cascaded classifiers) under distribution shift; mathematics of neural networks; applications to agriculture and health in West Africa.

Research experience

Research Assistant (unpaid), safety-constrained deep learning Oct 2023 – present
African School of Economics; supervised by Dr Songbian Zimé

- First author of **CascadeCropNet**, a two-stage cascade for crop damage classification with a calibrated recall floor: 23,804 georeferenced images from Kenyan smallholder maize farms; damaged-class recall about 0.97 on held-out data; 13.2% fewer expert calls. Under peer review, *Computers and Electronics in Agriculture* (arXiv:2607.25468).
- Open problem: calibration and test splits are random, not spatial, so the recall floor carries no guarantee outside the calibration distribution. A literature review on the recall floor as a probability constraint (Oct 2026) maps the gaps, the hardest being shift with little information about the target.
- Second application, **MalariaCascade** (with Michel Emel): the same design on 27,558 NIH malaria cell images, about 96% recall on clean images while a single network falls to 32.8% under heavy sensor noise. Preprint arXiv:2609.33005.

Research Assistant, climate-informed maize yield forecasting Jul 2026 – present
African School of Economics; supervised by Dr Jean-Claude Ahouangbenon

- Commune-level panel for Benin, Burkina Faso and Togo (155 communes, 1984 to 2023, about 217,000 records). First rainfall-only results were negative; adding vegetation data.

Research Assistant (data scientist), Aguda digital history platform Sep 2026 – present
African School of Economics; with Prof. Leonard Wantchekon (ASE / Princeton) and Dr Guilherme Lambais (University of Porto)

- Audited six archival datasets on Africans returning from Bahia to West Africa (3,399 passport records, 762 Ouidah baptisms, a 1,254-name baptism network). Wrote a Python checker that recounts all 426 figures in the project documents from the source files.
- Reproduced the team's record linkage (same 100 name pairs, 139 returnees) and documented 58 data and code issues with locations and fixes.

External collaborator, pretext-translate (open-source translation pipeline) Apr 2026 – present
IDEMS International

- Audited two corpora, found and fixed two high-severity extraction bugs, classified all 315 PreTeXt schema elements, wrote the audit report and a French onboarding guide.

WIIRU Learn, adaptive tutoring with Bayesian Knowledge Tracing 2026 – present
Supervised by Dr Yaé Ulrich Gaba

- Working paper for technical-track Terminale students in Benin. Not deployed.

Publications and manuscripts

- **Hagbe, J.T.M.**, Gounou, G.K. and Zimé, S.K. (2026). Safety-Aware Cascaded Inference for Crop Damage Assessment with Controlled Error Trade-offs. arXiv:2607.25468 [cs.CV]. Under peer review, *Computers and Electronics in Agriculture*; SSRN 7237146.
- **Hagbe, J.T.M.** and Emel, M. (2026). Safety-Constrained Cascade Inference for Robust Malaria Cell Classification Under Field Corruptions. Preprint, arXiv:2609.33005.
- **Hagbe, J.T.M.** (2026). Recall guarantees for cascaded classifiers under distribution shift: a positioning review. Manuscript, October 2026.
- Co-author of two mathematics textbooks adopted nationally in Benin (Collection Les Palmiers, 2024; Decree No 053/MESTFP).

Teaching

Teaching Assistant, Econometrics and Linear Algebra Oct 2023 – Jul 2025
African School of Economics, MSc Mathematics, Economics and Statistics

Mathematics Teacher, Terminale classes current
Complexe Scolaire Sainte Bakhita

Mathematics Teacher and Team Coordinator (part-time) Oct 2019 – Dec 2024
Complexe Scolaire Pierre Joseph de Clorivière, Abomey-Calavi; baccalaureate classes, programme planning and support to colleagues

Member, national baccalaureate mathematics grading committee 2019 – present
Republic of Benin

Mentor, time series forecasting in R Nov 2025
AFRICAST workshop, Cardiff University

Education

MSc Mathematics, Economics and Statistics 2022 – 2025
African School of Economics, Abomey-Calavi (taught in English). GPA 3.61/4.3

- Thesis (defended 25 Sep 2025): Digital Financial Services and Poverty Reduction in Benin; multivariate probit on Global Findex 2021 and interviews with 34 women in four communes. Supervisor: Segnon Aguey, PhD.

Master in Pure Mathematics and Applications 2018 – Jan 2022
Université d'Abomey-Calavi. Second Class Honours (Upper Division)

- Thesis (defended 22 Jan 2022, 15.5/20, Very good): neural network regression in high dimension, an expository study of Bauer and Kohler (2019) with original numerical simulations.

Licence in Mathematics and Applications 2014 – 2018
Université d'Abomey-Calavi. Benin Government Scholarship (2016 – 2018)

Training and awards

- Selected participant, Applied Mathematics Foundations of AI Summer Academy (fully funded), MCAIM, University of Michigan, with JKUAT, Nairobi, Aug 2026.
- AIMS RIC Doctoral Training School, Foundational Methods in Data Science, Kigali, Mar – Apr 2026: Excellence Scholarship; Best Participant Award (peer-selected); research presentation, 20 Mar 2026.
- Shortlisted, AI, Uncertainty & Simulation Workshop (Imperial College London, Mauritius), 2026.
- Africa TechUp programme (ISHEERO and 54Bridges, 2024); QuantEcon Africa (ASE, 2023); IBM Data Science Specialization (2023 – 2024).

Skills and languages

Python, PyTorch, R, SQL, Stata, LaTeX, Git. French (native), Fon (native), English (professional working proficiency).

References

Available on request: Prof. Leonard Wantchekon (Princeton University and ASE); Dr Songbian Zimé (ASE); Dr Yaé Ulrich Gaba (Sefako Makgatho Health Sciences University); Prof. Guy A. Degla (IMSP, Université d'Abomey-Calavi).